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Spatiotemporal Multi-objective Optimization for Competitive Mobile Vendors' Location and Routing Using De Facto Population Demands

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Mobile vendors such as food trucks have become viable players in the food service market as a relatively new business model based on high mobility and flexibility. To remain competitive against typical brick and mortar establishments, it is important for mobile food vendors to operate at sites that will generate sufficient revenue, which may mean navigating to locations of high demand and low competition. This study addresses the optimal locations and routing of mobile vendors along different objectives that could contribute to higher revenue. We develop a multi-objective spatiotemporal optimization of food trucks by explicitly considering the dynamics of urban population distribution and thus potential customer demand. For an empirical application, we focus on the location and routing problems of food trucks in Seoul, Korea. The results show that several noninferior solution sets are possible for simultaneously maximizing demand capture while reducing market competition. A routing between optimal locations for food trucks is also suggested. This research could provide useful insights and practical solutions that are directly applicable to various mobile vendors.

Introduction

Central place theory helps to explain the historic need of buyers and sellers to meet at a central market (Christaller 1966). In this regard, mobile vendors have played a significant role in serving particularly in sparsely populated areas; however, their role in current markets is evolving. In developing countries, mobile vendors such as food trucks have become significant sources of economic prosperity for vendors and food provision for customers. Unlike traditional mobile businesses, mobile vendors represent a new business model in that they can adapt to moving and potentially unmet demand, taking into account existing competitors already positioned in the area.

The success of mobile vendors in the market could be aided by knowing the spatial distribution of human movement in urban areas. Traditionally, potential population and customer

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demand can be estimated from residential areas and census tracts (Satti and Jacobs 2004; Ni et al. 2005; Tong and Murray 2009). Today, spatiotemporal variability of actual demand is augmented by public transportation and personal mobility data. With individual mobility data, which can be digitally stored and traced from personal mobile phones, mobile vendors can increase sales by moving to locations of dense customer demand at specific times of the day. To represent dynamic changes in the human movement for mobile vendor location modeling, real-time population distributions (de facto population) rather than the settled population (de jure population) should be used.

In this study, we address optimal locations of mobile vendors by relying on actual demand distributions over time based on the de facto population derived from hourly mobile phone data. Particularly, we highlight food truck sale locations in Seoul, Korea. High-resolution data for spatiotemporal distributions of the population are readily available, as is spatial information on mobile vendors such as food trucks. It is expected that the proposed multi-objective model will produce solutions that give the locations at which for food trucks can efficiently cover demand at all times. More practically, food trucks can provide their service to areas of uncovered demand, such as transient food deserts or areas with sporadic population density, such as business districts.

Up until recently, changing locations for food trucks was not legal in Seoul. Before enacting the ordinance of code 7,782 in April 2017, food trucks could only stay in designated places without moving. Therefore, food trucks could only serve a limited number of customers within a permitted zone specified by the law. However, commercial activities of food trucks has accelerated by two significant deregulations in recent years. First, food truck vendors have been allowed to change their business locations during the workday. Second, possible business areas for food trucks have been lawfully expanded. Now, food trucks can move to maximize customer demand, and need to consider the optimal routes between business spots. Simultaneously, food trucks need to avoid the expected competition of existing mobile and established retailers as possible. Considering both concerns of food trucks, we develop a multi-objective optimization model which has two conflicting objectives, which is to minimize the competition between existing retailers and new mobile vendors while maximizing the demand captured. In our proposed model, a greedy adding algorithm is suggested by combining a competitive location model which is formulated as an integer program. The specific research flow is presented in Section “Methods for optimal food truck locations and temporal routing”.

This article is structured as follows: In Section “Literature review”, we briefly review previous studies of food truck location models and spatio-temporal analysis. The methodology of this study is introduced in Section “Methods for optimal food truck locations and temporal routing”. The empirical application of food truck location modeling is presented in Section “Empirical application of food truck business in Seoul”. Section “Conclusions and discussion” summarizes our results and discusses further research.

Literature review

The earliest study of the periodic market comes from “The Isolated State”, by Johann Heinrich von Thünen (1966), but Christaller (1933) explains the periodic market as an important step for constructing the permanent market structure. When low population does not meet the threshold for sufficient demand, suppliers must move to areas with greater demand. Thus, suppliers continue mobile economic activities until the minimum threshold is reached within a maximum range of service. In the empirical context, Stine (1962) focused on the periodic markets in Korea.

By investigating workplace and worker routes in the periodic market, he attempted to explain periodic market characteristics based on the range and threshold concepts of the Central Place Theory. Stine argued that if the range of the goods is smaller than the threshold, then the trader becomes a mobile vendor. Mobile vendor locational behavior was modeled with a mathematical approach by Ghosh (1982). Gosh analyzed how to maximize total profits of periodic market workers by applying an optimization approach. He focused on the locational strategy and the consequent spatiotemporal pattern of the periodic market system.

Previous studies have focused mostly on the development and characteristics of the periodic market. A mobile vendor such as a food truck has better mobility than traditional periodic market vendors and represents a new model for analysis beyond serving as an intermediate stage toward a permanent market as argued by Christaller. Mobile vendors and existing competitors with fixed stores in the same market area may lead to a hybrid market structure that allows periodic and permanent markets in the same area. Mobile vendors might make compromises to avoid competition with existing merchants as much as possible.

As a mobile vendor, the food truck is an emerging business model in Korea, and there are a few studies regarding food trucks. Most research has focused on the food truck business model as a restaurant (Joo 2018; Lee 2018) and related regulation problems in Korea (Kim 2014; Park 2015; Moon 2017). Other studies have discussed the value of food trucks in terms of food deserts, inequality of the food supply (Acho-Chi 2002), and the enhancement of public health (Wallace 2012; Lasserre 2013). In addition, food truck research related with social media and smartphones were also explored (Esparza, Walker, and Rossman 2014). Although some studies addressed the issue of food truck routing (Liu 2013; Bhandawat 2018), the efficient location of food trucks and competing for market environments have not been explicitly discussed.

Furthermore, changing temporal demands in actual markets for food truck location modeling has not been considered. Population flow in urban areas over time may be a good representation of the temporal dynamics of demand. With the advance of telecommunication technology and the high penetration rate of mobile phones, temporal distributions of population can be explored based on mobile phone location data. For example, Deville et al. (2014) mapped a dynamic population using mobile phone location data to reconstruct a more exact population distribution in Portugal. Similarly, mobile call data can be used to estimate temporal urban population distributions (Kang et al. 2012). Moreover, spatiotemporal human mobility patterns can be examined with mobile phone data (Gao 2015). Despite those efforts, it is still challenging to use mobile phone data from commercial telecom companies in many countries due to regulation and expensive costs. In this study, we use different temporal population distributions estimated from mobile phone data and explore optimal locations for food trucks in particular time periods. To consider the competitive market environment among mobile vendors and existing merchants, and to model their conflicts appropriately, we develop a multi-objective location model. There are two primary objectives with trade-offs. Mobile vendors usually wish to maximize their profits, while they attempt to avoid unnecessary competition with existing retailers at a fixed business location. The optimized solutions are examined with regard to different importance for each objective. There has been a number of studies on the spatial multi-objective optimization for various applications, which seek to a set of solutions that is not inferior to any other feasible solutions (Jaszkiewicz and Czyzak 1998; Kuby et al. 2005; Ligmann-Zielinska, Church, and Jankowski 2008; Lee and O'Kelly 2009; Cao et al. 2011).

Methods for optimal food truck locations and temporal routing

This study integrates multiple spatial analytics through three major phases. In the first phase, we define the service area of mobile vendors by applying the competitive location model. A multi-objective approach based on spatial optimization is introduced in the second phase. The greedy algorithm is used in the solution procedure. In the final phase, data mining techniques and the vehicle routing problem (VRP) are deployed to define the food truck routing sets. Because the VRP is a combinatorial optimization problem and thus it is not easy to be solved in a polynomial time, a heuristic method has been deployed for solving the VRP. In this article, we use the savings algorithm suggested by Clarke and Wright (1964), which has been widely adopted for solving the VRP due to high optimality (Agarwal, Mathur, and Salkin 1989). After solving the VRP for food trucks, optimal configurations of food truck locations are derived by explicitly considering existing restaurants and temporal population distributions.

Service area and customer allocation

The competitive location problem has been studied since the seminal work of Hotelling (1990). In this study, food truck demand is defined with a deterministic competitive model considering two restricted assumptions for the probabilistic model: small service range and homogeneity of food truck service (Huff 1964; Nakanishi and Cooper 1974; Stanley and Sewall 1976; Eiselt and Maranov 2011; Lee and O'Kelly 2011).

For the deterministic location model, square grids are formed to calculate the range of food trucks. A food truck captures all of the population in the range (the black circle in Fig. 1). The service range of a food truck is defined as 200 m from the centroid of the grid. With less attraction than typical restaurants, people might tend to spend less than 5 minutes to access to a food truck showing a relatively short service range (Jin 2017). Therefore, the service range of a food truck is reasonably assumed as 200 m from its location for the food truck's allocation problem here.

The competitive market model shares demand when ranges overlap. In this study, food trucks have their own range and attraction. When two food truck ranges are overlapped, the overlapped demand is allocated to the closest food truck. As shown in Fig. 1, the overlapped area can be divided into two parts. If the food truck is located at O_i and the range of food truck at O_n is overlapped, the demand population P_n captured by food truck at O_n is re-calculated as follows:

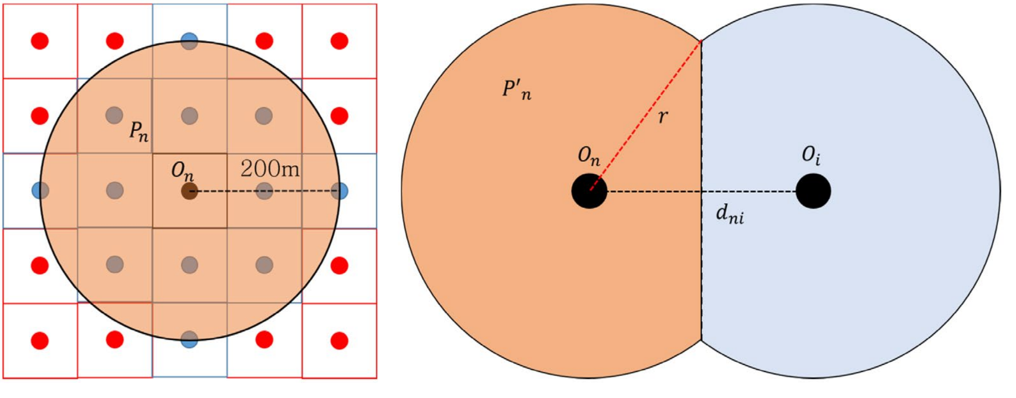


Figure 1. Food truck service range and demand allocation.

$$P'_n = \left\{ A + \left[\left(\frac{d_{ni}}{2} \times \sin\left(\cos^{-1}\left(\frac{d_{ni}}{2r} \right) \right) \right) \times r \right] - \left[\left(\pi r^2 \times 2\cos^{-1}\left(\frac{d_{ni}}{2r} \right) \times \frac{1}{2\pi} \right) / A \right] \right\} \times P_n \tag{1}$$

where P'_n is the demand population recalculated due to the overlapped market share from the competitor; A is the area of the range of food truck at O_n; d_{ni} is the distance between two food trucks at n and i; r is the range of a food truck at O_n. P'_n is derived by subtracting the portion captured by the competitor from the entire circular market area. By calculating P'_n, the demand, which can access multiple facilities would not be counted multiple times.

Multi-objective optimization model for food truck locations

Optimal locations for food trucks are explored in a multi-objective optimization framework. Generally, multi-objective optimization is applied when two or more objectives are considered simultaneously. Usually, all objective functions are combined into one single objective with a weighting method (Daskin 2011). For the optimization problem in this study, there are two conflicting objectives. The first objective is to maximize the capture of potential demand in each time period. Another objective is to minimize competition among existing restaurants and other food trucks. The customer capture for each food truck is based on the de facto population within the service range. To measure the degree of competition in the market, we define the competition index by calculating the number of established restaurants within the service range of a potential food truck location. Both the capture of demand and the degree of competition are Z-score normalized for calculating the entire objective function. Equation (2) defines the capture of customers and equation (3) defines the degree of market competition for a particular time period and area (grid zone). Equation (4) shows a single objective function in a multi-objective optimization framework by combining two objectives of equations (2) and (3) with the weight α. A larger α indicates greater importance of customer capture than avoidance of competition within the market. Equation (5) restricts the total number of food trucks to be located for the particular time period. The capture of potential demand by new food truck location at a particular time period t in the grid zone i is calculated with equation (6). Equation (7) defines the integer constraints of food truck location and feasibility for siting in business in a particular time period t and grid zone i.

$$\text{Maximize } Z_1 = \sum_t \sum_i I_i P'_{ti} X_{ti} \tag{2}$$

$$\text{Minimize } Z_2 = \sum_t \sum_i N_i X_{ti} \tag{3}$$

$$\text{Maximize } Z = \alpha Z_1 - (1 - \alpha) Z_2 \tag{4}$$

Subject to

$$\sum_i X_{ti} = F \quad \forall t \tag{5}$$

$$P'_{ti} = \left\{ A + \left[\left(\frac{d_{ni}}{2} \times \sin\left(\cos^{-1}\left(\frac{d_{ni}}{2r} \right) \right) \right) \times r \right] - \left[\left(\pi r^2 \times 2\cos^{-1}\left(\frac{d_{ni}}{2r} \right) \times \frac{1}{2\pi} \right) / A \right] \right\} \times P_{ti} \times X_{ti} \quad \forall t, i. \tag{6}$$

$$X_{ti} = \{0, 1\}, I_i = \{0, 1\} \quad \forall t, i \tag{7}$$

where,

$$X_{ti} = \begin{cases} 1 \text{ if food truck is located in grid } i \text{ at period } t \\ 0 \qquad\qquad \text{otherwise} \end{cases}$$

$$I_i = \begin{cases} 1 \text{ if grid } i \text{ is feasible in business for food trucks} \\ 0 \qquad\qquad \text{otherwise} \end{cases}$$

P_{ti}, the de facto population at period t in grid i;

P'_{ti}, the de facto population captured by new location of food truck in grid i at period t;

N_i, the number of restaurants in the range of grid i; F, the number of mobile vendors; r = the food truck service range; A, the area of food truck service range;

d_{ni}, the distance between two food trucks at n and i sharing a service area.

Optimizing the routes of food truck temporal movement

Although food trucks can survive with low demand, they will need more customers to sustain their business. Most customers patronize established restaurants; mobile vendors may only capture irregular customer flows. Accordingly, mobile vendors often move between several places to serve as many customers as possible. Generally, the cost of moving to different sites is manageable. However, for more cost-effective travel between sites, optimizing the routes of food trucks for changing demand distributions overtime is required. According to an interview with food truck workers, changing locations frequently is undesirable because of the fuel expense and the effort in setting up in a new location. Food trucks tend to maintain their location unless there is a significant decrease in demand (Moon 2017). In this study, we define a 20% reduction in the population for a particular time period as a threshold for food truck movement. Based on the interview, a 20% reduced demand is sufficient motivation for changing locations for food trucks due to their terminal cost. Food trucks that move to another site can be defined by the following conditions.

$$X_{ti} > X_{(t+1)i} \tag{8}$$

$$0.8 \times P_{ti} > P_{(t+1)i} \tag{9}$$

The inequality (8) indicates that the mobile food truck location contains a smaller number of food trucks in cell X_i in the period $(t + 1)$ than in period t. When the number of food trucks in the cell decreased, the optimal routes for moving were suggested. The condition (9) indicates that food trucks move when the de facto population at time period t in grid i (P_{ti}) decreases by more than 20% at time period $t + 1$ (P_{t+1i}). The selected food trucks at the time period t are allocated to new optimal locations at the time period $t + 1$. For developing the optimal routing of food trucks, we used the VRP framework, which was first presented by Dantzig and Ramser (1959). The VRP is the optimization of delivery or collecting routes from several points to a number of scattered sites to minimize total travel costs (Laporte 1992). Because the food truck capacity is unlimited, only the distance condition is imposed. Different time periods are considered for moving routes.

Empirical application of food truck business in Seoul

For an empirical application, we examine the food truck sites and routes operating in Seoul, Korea. The Seoul metropolitan government is addressing food truck locations as an important urban issue; over 35% of food trucks go out of business within six months after opening (Seoul Metropolitan Government 2017). Therefore, we expect that our spatiotemporal optimized solutions using actual hourly population data provided by the Seoul metropolitan government will be a useful support for food truck survival in the market.

Study area and data

Seoul consists of 25 *Gu*, 424 *Dong*, and 19,153 *Jipgyegu*[1] as administrative units. With the availability of hourly de facto population data, spatial distributions of the population with higher temporal resolution can be developed to represent dynamic demand. As food trucks are small businesses, demand elasticity is significant, and it is necessary to analyze nearly real-time population flow. The de facto population is derived from KT[2] telecommunications big data and residential census data from the Seoul metropolitan government and KT.

Mobile phone data are an effective source to represent the current population at a particular place (Gao 2015; Lee and Kim 2016). Moreover, various factors affecting spatial decisions on the movement by food trucks should be also considered, for example, road networks, regional characteristics, and legal regulations. The number of food trucks to be located (F) is defined as 500, the same as the current number of food trucks in Seoul.

The potential areas for food truck locations are predefined by the Seoul Metropolitan Council. Several land uses are allowed for the legal operation of food trucks (festival, public parking lot, public facility, traditional market, plaza, public property, and roadside). By considering limited areas for the food truck business, the computational burden of the model can be reduced. For creating the candidate sites, we tessellate the whole area of Seoul with a finer-resolution grid with 100×100 m cell. The resulting number of candidate sites is 61,650. In the model, the total number of feasible zones for siting was reduced from 61,650 to 44,714. Nevertheless, with a finer grid set of potential zones, it is still difficult to solve the problem in a reasonable amount of time because of the heavy computation load, $\binom{44,714}{500}$, for four time periods. Therefore, the proposed multi-objective optimization model is solved using a greedy adding heuristic algorithm.

Optimizing food truck locations

Fig. 2 shows the general pattern of the de facto populations of weekdays over time in Seoul. The maximum de facto population per day in Seoul is greater than 10,950,000; the minimum population is less than 10,650,000. After 22:00, most commuters to Seoul metropolitan areas return to their homes, and only the resident population remains.

Fig. 3 presents the spatial distributions of the population in the morning (6:00 am) and evening (6:00 pm). The population at 6:00 am means residents who are still at home. Major residential areas of Seoul are highly populated in the left side. The weekday distribution of the de facto population illustrates a typical spatial separation or mismatch between the residential area and the business district. Seoul has multiple employment centers that form a polycentric urban structure. In the de facto population distribution, urban centers and CBD areas are densely populated at 6:00 pm (i.e., circled areas of *Jongro* and *Gangnam*) but are sparsely populated at 6:00 am. This shows the spatial dynamics in Seoul.

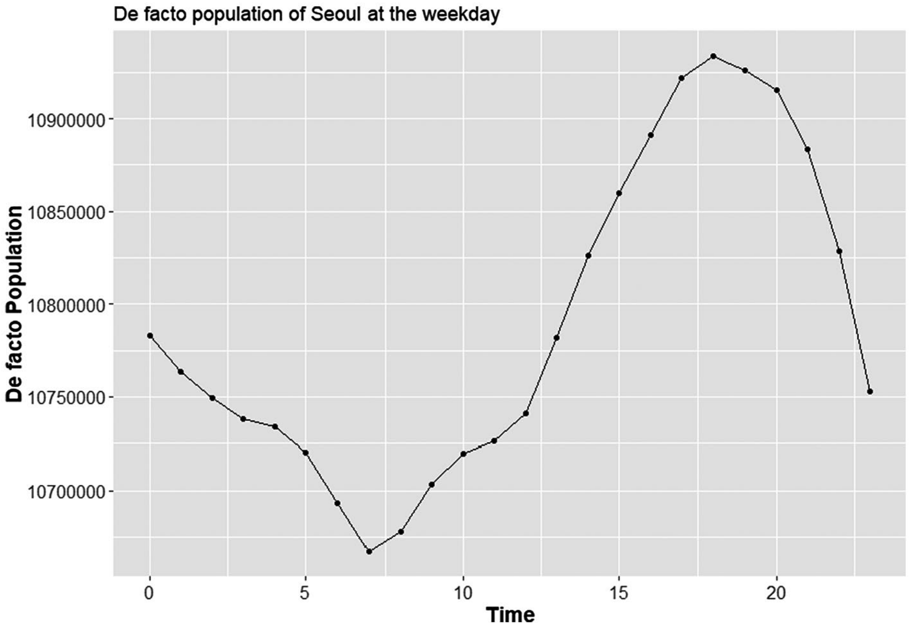


Figure 2. De facto population of Seoul.

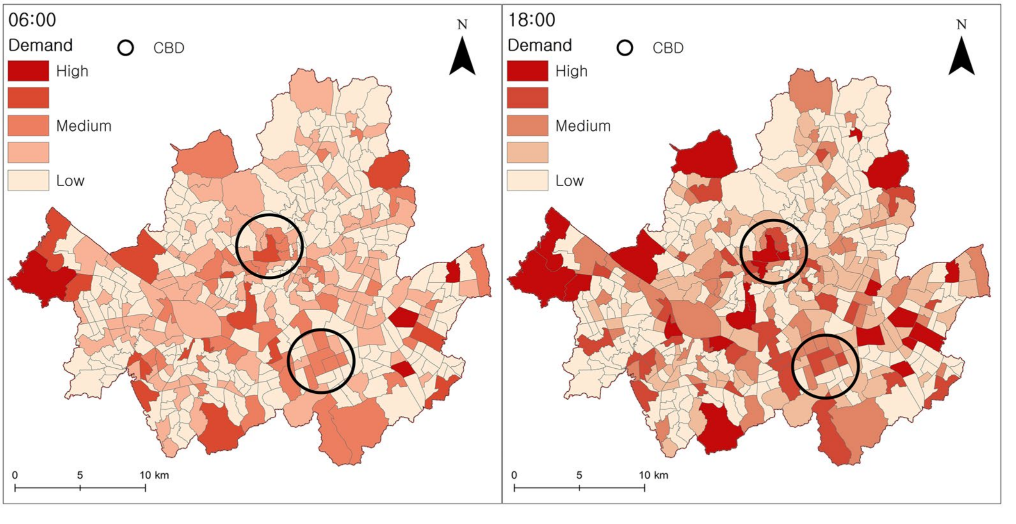
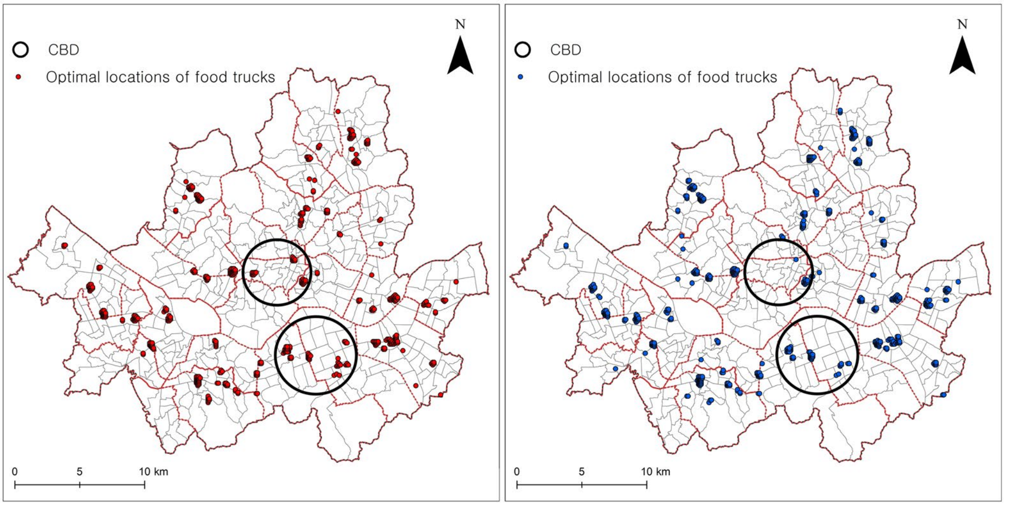


Figure 3. Population distributions at 6:00 am (left side) and 6:00 pm (right side).

With a framework of multi-objective optimization, several solutions can be simulated using different weights of demand capture and competition avoidance. Table 1 presents the results of objective functions with different weight factors. As the weight factor α increases, the demand captured by food trucks and the number of restaurants in competition increase. Therefore, a high weight factor and emphasizing captured demand make the market more competitive by inviting more players. For example, when α is 0.9, the customers captured by food trucks double, with

Table 1. Objective Function Results in Terms of Different Weights for Multi-objectives

Weight factor (α)	Number of restaurants in competition	Mean number of restaurants in competition	Demand captured	Mean demand captured
0.1	4,644	9.288	45,183,106	90,366
0.3	36,933	73.866	64,315,346	128,631
0.5	91,040	182.08	77,816,843	155,634
0.7	182,525	365.05	84,741,657	169,483
0.9	281,730	563.46	88,420,654	176,841



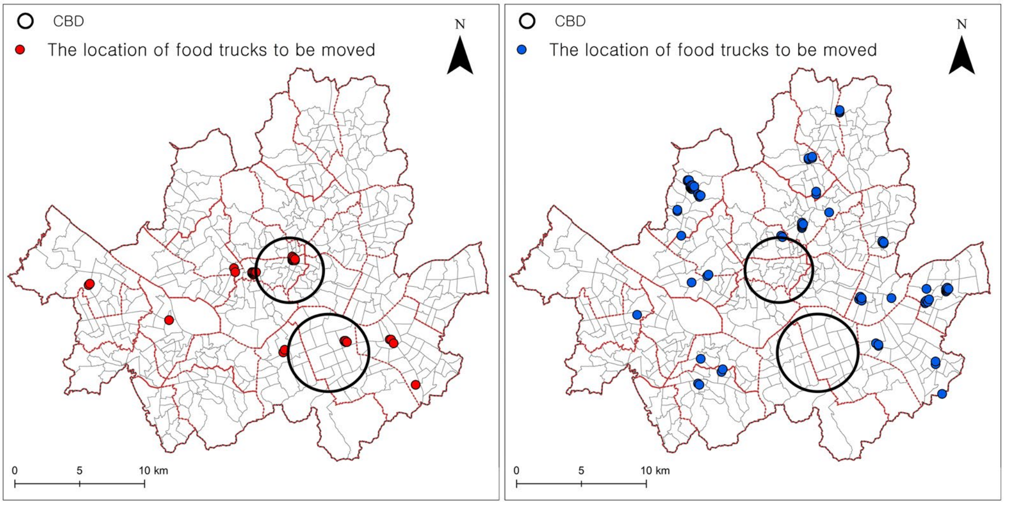
(a) Lunchtime (11:01-13:00) **(b)** Dinnertime (17:01-20:00)

Figure 4. Optimal locations of food trucks for different weekday time periods.

more than 563 established restaurants in their market area. In contrast, when α is 0.1, the average capture by a food truck is 90,366 customers. However, only 9.2 restaurants will be affected by one new food truck.

An α value of 0.5 indicates equal weight for market competition and demand capture. Fig. 4 shows the optimal locations of food trucks for different weekday time periods when α is 0.5. Food trucks are usually located in major city centers at lunchtime, Fig. 4a) because many workers are in the business and commercial districts during weekdays. Although food trucks compete with many existing restaurants, they are willing to be positioned in the CBDs and nonresidential areas (circled areas in Fig. 4) because of large potential demand. At dinnertime, Fig. 4b), food truck vendors move to the outskirts of Seoul or near residential areas based on changing demand distribution. In the both time periods, clustered food truck sites are presented with points.

Fig. 5 shows that food trucks move after each time period. After lunchtime on weekdays, the food trucks in CBDs need to change their location for dinnertime in Fig. 5a). Because the residential areas are on the outskirts of Seoul, the food trucks operate there at dinnertime to capture the residential population. After dinnertime, in Fig. 5b), the food trucks gather again in the CBDs from the outskirt of Seoul. This spatial behavior represents the spatiotemporal dynamics of food truck locations in Seoul.



(a) After lunchtime **(b)** After dinnertime

Figure 5. Optimal locations of food trucks for different weekday time periods.

Model validation

The results of the research are compared with current food truck locations to validate our model. Fig. 6 illustrates that the current food truck locations are inferior to the optimized solutions for both objectives. Because all of the four time periods show very similar graphs, Fig. 6 was created by the sum of four time periods. In the graph, the blue dot (the current condition) is located outside the trade-off frontier, indicating that food trucks can capture more customers and reduce conflict with existing restaurants by operating with weight factors of 0.1 and 0.3.

The nondominated solution sets from research present noninferiority than the current locations of food truck. When the weight factor for the demand is less than or equal to 0.3, total number of competing restaurants within the service areas by newly sited food trucks (36,393 restaurants for the weight of 0.3) is smaller than the number of restaurants competing with currently sited food trucks (50,375 restaurants competing against current food trucks). It means that the solutions for the weights less than or equal to 0.3 implicate less competitive scenarios between existing restaurants and new entry of food trucks. As the weight for the demand increases, the population captured by new food trucks increases, and the competition between existing restaurants and food trucks gets severe (Table 2).

For a clear visual representation, the K-means clustering method is applied to represent the food truck location change. After detecting nine clusters to group similar food truck locations by their silhouette score, the centroid of each cluster is regarded as a representative business spot for a group of food trucks. Fig. 7 shows the food truck cluster centroids at lunchtime and dinnertime on the weekend. At lunchtime, the clusters are located in the CBDs; they move away from the CBDs before dinnertime. At dinnertime, the clusters are mostly in residential or outer areas of the city.

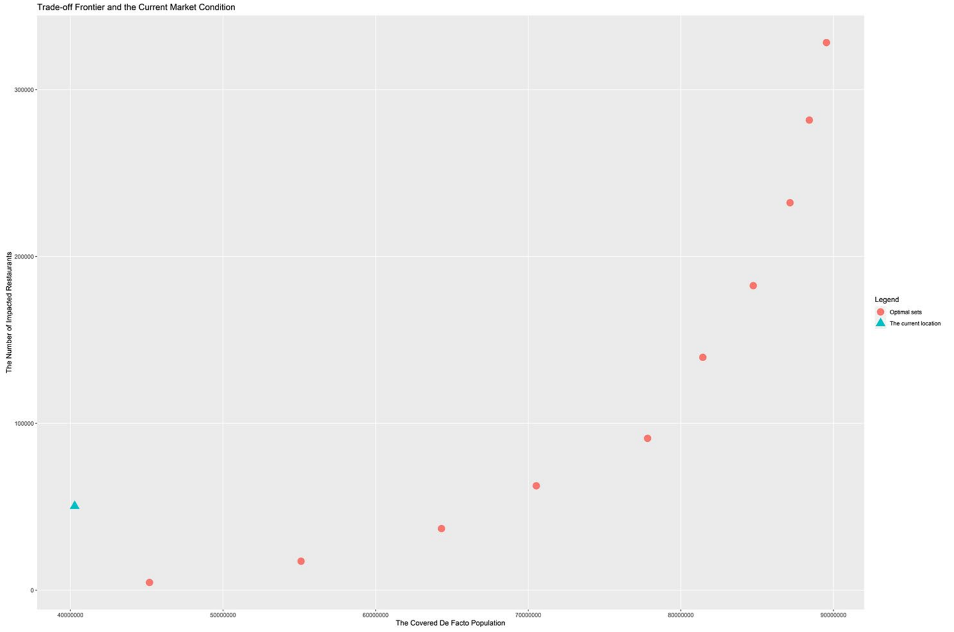


Figure 6. Comparison of the trade-off frontier for optimally chosen locations and current location for food trucks.

Table 2. Comparison of the Current Market Conditions and Optimized Solutions

Current market conditions		Optimized solutions		
Number of restaurants	Demand captured	Weight factor	Number of restaurants	Demand captured
50,375	40,276,752	0.1	4,644	45,183,106
		0.3	36,933	64,315,346
		0.5	91,040	77,816,843
		0.7	182,525	84,741,657
		0.9	281,730	88,420,654

Fig. 8 shows the space–time prism of five representative mobile food trucks for four time periods. Specifically, it represents the spatiotemporal optimal flows of food trucks for each time period from weekday lunchtime to weekend dinnertime. It is known that a space–time prism is an effective tool for visualizing the spatiotemporal flow (Egenhofer and Golledge 1998; Kwan and Hong 1998; Miller 1999). It seems that the clustered locations of mobile food trucks are changed from the CBD at lunchtime to the outskirts of Seoul at dinnertime. Eventually, food trucks return to previous optimal locations at weekday lunchtime.

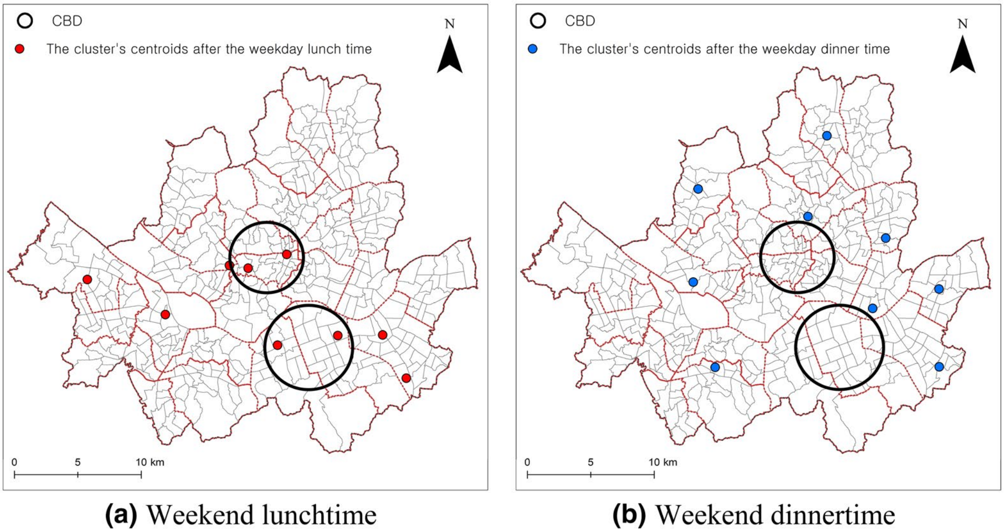


Figure 7. Cluster locations of mobile food trucks for different time periods.

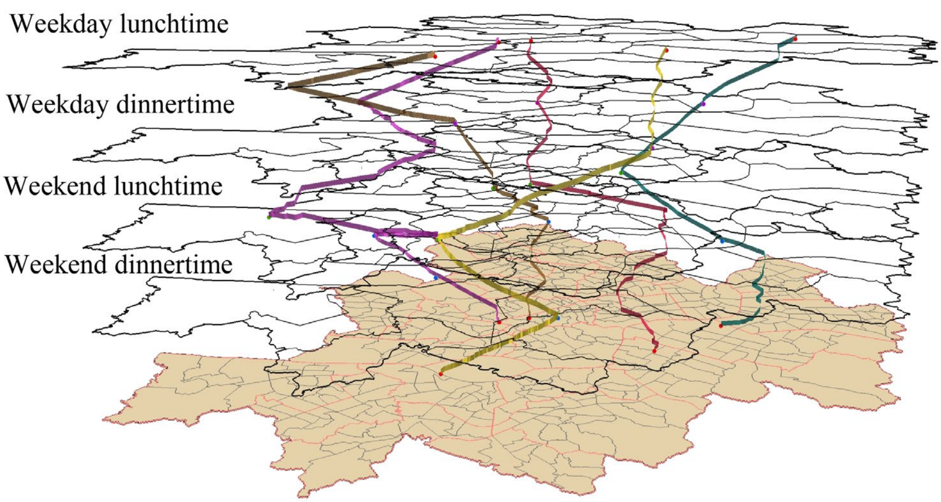


Figure 8. Spatiotemporal optimal routes of food trucks. The video can be accessed at the following address: https://drive.google.com/file/d/15nQ5gNbkHRzaxOQEqjaYwjmJVno1RTSo/view?usp=sharing.

Conclusions and discussion

Before the advance of transportation and telecommunication technology, a mobile vendor was an important actor in the periodic market, providing necessary goods in sparsely populated areas. Today, mobile vendors are competitive players in the mature market with new mobile businesses such as food trucks. They can enter into the mature market serving uncovered demand from established merchants. Their success depends on their mobility in responding to actual demand distributions that vary over time. This study addressed temporally changing optimal locations and routes for mobile vendors. We developed a multi-objective spatiotemporal optimization model for food trucks considering actual demand distribution at a particular time in a particular area. In the optimization process, the objectives of customer capture and reducing competition were systematically simulated with different scenarios.

For an empirical application, we applied the mobile vendor location and routing problem to the food truck business in Seoul, Korea. The dynamics of demand distribution were modeled by the de facto population estimated from mobile phone data. Using the model, we solved the problem more efficiently with a greedy heuristic method. Several sets of nondominated solutions were presented with different scenarios of optimization. The trade-off between demand capture and the competition was clearly observed. As the demand capture is emphasized, more existing merchants are in competition with food trucks. Different sets of food truck locations were optimally selected for each time period. The optimized solution sites varied according to the distribution of the de facto population for each time period, typically showing the polycentric characteristics of Seoul. The routes to optimized food truck locations over time were also suggested. Such optimal routes for mobile vendors best locations for four consecutive time periods are obviously useful for efficient scheduling for best sites for sale.

The results of this research could be used by mobile vendors to gain useful insights and practical solutions that can be directly applied to their business. Further, scenario-based solutions could help make flexible decisions. It is also worth noting that this research shows how the de facto population-based on mobile phone data can be used in location modeling and support for more practical solutions. However, it is still a challenge to more effectively visualize solutions that are temporally changing. This research requires a number of analytical processes to solve the problem. In future research, a combined framework with spatiotemporal location optimization and routing optimization could be developed.

Acknowledgments

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Notes

1 Jipgyegu is the minimum census unit in Korea. Mean residential population and area per Jipgyegu is 643 and 0.03 km^2. This research used Jipgyegu as an areal unit for the analysis.

2 Telecommunication company in South Korea.

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